Equilibrium and missing forces

antocciphysics | TEACHER KEY | CPA

Learning target: Find a missing force when acceleration is zero.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 15; 3 course days later.*
Only weight 30 N downward from Earth. The hand is no longer in contact; upward velocity is not a force.

02. Explain a worked example (worked)
4+6-10=0 N; upward total equals downward total.

03. Find the balance (model)
Friction 20 N left; net 0 N.

04. A still lamp (practice)
30 Nup.

05. Parked or cruising? (reason)
First two can; turning car has changing velocity and nonzero net force.

06. Choose a method (mixed) *Review: Unit 1, Day 6; 12 course days later.*
3 m east and 4 m south. Add each pair of signed components.

07. Exit check (exit)
Additional 7 N left; 12-5-7=0 N.

08. STRETCH: Apply it another way (stretch)
Examples (0, 10),(2, 8),(5, 5) N; sum remains 10 N.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 15; 3 course days later.*
Weight down, normal up, kinetic friction right. No vertical acceleration; rightward horizontal acceleration if friction is the only horizontal force.

02. Explain a worked example (worked)
4+6-10=0 N; upward total equals downward total.

03. Two component balance (model)
(-12,-9) N; magnitude 15 N; 36.9° south of west.

04. Two-rope hanger (practice)
2 T sin 45=39.2; T 27.7 N. Only vertical components add up to weight.

05. Change angle (reason)
T 30=98/(2×.5)=98 N; T 60=56.6 N. Steeper ropes provide more vertical component per unit tension.

06. Choose a method (mixed) *Review: Unit 1, Day 6; 12 course days later.*
R=(2, 4) m; magnitude ≈4.472 m at 63.43° north of east. C=(-2, -4) m.

07. Exit check (exit)
T 29.4 Natrest; upward a requiresT-mg=ma, soT>29.4 N.

08. STRETCH: Apply it another way (stretch)
(-30,+40) N; magnitude 50 N; 53.1° above leftward horizontal. Balance must hold in both axes.